

EXCEL FOR DATA ANALYSTS

Categorized Theory Notes & Practical Reference

Built for interview preparation — with a Data Analyst lens on every topic

12 categories · PivotTables to VLOOKUP · relevance-rated for analysts

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Relevance rating tells you how central each topic is to a Data Analyst role: **HIGH** = use constantly, **MEDIUM** = useful, come back to it.

How to use this document: work through categories in order. Rebuild every example in an actual spreadsheet (not just read it), then read the 'Data Analyst angle' note — many Excel features map directly onto the SQL and Python concepts in the earlier notes, which makes all three easier to remember together.

01 Excel Basics — Interface & Ranges

HIGH

Before formulas and functions, you need to speak Excel's basic language: cells, ranges, and how the grid is addressed. Every formula you'll ever write depends on this.

CONCEPT / FUNCTION	WHAT IT DOES
Workbook / Worksheet	A workbook is the whole file; a worksheet (tab) is one grid of cells inside it
Cell Reference	A cell's address, e.g. B4 = column B, row 4 — how you point to any single value
Ranges	A group of cells, e.g. A1:A10 (column) or A1:D1 (row) or A1:D10 (block)
Excel Syntax	Every formula starts with = , e.g. =A1+B1 — Excel evaluates anything after = as a calculation
Selecting Cells	Click-drag, Shift+click, Ctrl+click — how you build ranges and multi-selections

Practical Example

Formula bar:

=A1+A2

Meaning: add the value in cell A1 to the value in cell A2, and show the result in the cell where this formula is typed.

Range example:

=SUM(B2:B10) -> adds every value from B2 through B10

■ Data Analyst angle: this is the Excel equivalent of SQL's FROM/table concept — a range is 'the data you're working with,' and almost every function later in this document takes a range as its main argument.

■ Interview tip: know the difference between a relative range (A1:A10) and naming a range (Excel lets you name a block of cells, e.g. 'SalesData') — named ranges make formulas far more readable and come up in advanced-Excel interview questions.

02 Entering, Filling & Editing Data

HIGH

The everyday mechanics of building a spreadsheet — filling patterns down a column, moving data around, and undoing mistakes without breaking formulas.

CONCEPT / FUNCTION	WHAT IT DOES
Fill	Drag the fill handle to copy a value or continue a pattern (1, 2, 3... or Jan, Feb, Mar...) down a column
Double-Click to Fill	Auto-fills a formula down an entire column as far as the adjacent column has data — a major time-saver

CONCEPT / FUNCTION	WHAT IT DOES
Move Cells	Cut/drag cells to a new location; formulas referencing them adjust automatically
Add / Delete Cells	Insert or remove cells, rows, or columns, shifting surrounding data accordingly
Undo / Redo	Ctrl+Z / Ctrl+Y — essential safety net, especially before you're confident with formulas

Practical Example

Column A: Month names

A1: Jan

A2: (drag fill handle down) -> Feb, Mar, Apr, ...

Column B: a formula

B2: =A2&" report"

(double-click the fill handle on B2's corner)

-> auto-fills "Feb report", "Mar report", ... down as far as column A has data

■ Data Analyst angle: fast, correct filling is what separates someone who can build a 10,000-row model in 5 minutes from someone who manually types every row — this is a genuine speed skill interviewers watch for in live tests.

■ Interview tip: double-click-to-fill is a small but real productivity signal — mentioning it (and Ctrl+Shift+End to jump to the last used cell) shows you've actually worked in Excel daily, not just watched a tutorial.

03 Formulas, References & Operators

HIGH

This is the single most important category in the whole document. Relative vs. absolute references trip up more people in interviews and real work than any function name.

CONCEPT / FUNCTION	WHAT IT DOES
Formulas	Any calculation starting with = — can combine cell references, numbers, and functions
Relative Reference	A1 — changes automatically when you copy the formula to a new cell (adjusts to the new row/column)
Absolute Reference	\$A\$1 — locked; stays exactly the same no matter where you copy the formula to
Arithmetic Operators	+ (addition), - (subtraction), * (multiplication), / (division)
Parentheses	Control the order of operations, exactly like in math: =(A1+A2)*B1
Functions Overview	Pre-built formulas like SUM(), AVERAGE(), IF() — covered function-by-function later in this document

Practical Example

Relative reference (adjusts when copied down):

B2: =A2*10%

B3: =A3*10% <- automatically updated when copied from B2

Absolute reference (locked, e.g. a fixed tax rate in D1):

B2: =A2*\$D\$1

B3: =A3*\$D\$1 <- still points to D1, even after copying

Order of operations with parentheses:

=(A1+A2)*B1 vs =A1+A2*B1 <- these give DIFFERENT results

■ Data Analyst angle: forgetting an absolute reference is the #1 cause of 'why did my whole column break' bugs when copying formulas down 500 rows — get comfortable with F4 (the shortcut to toggle reference types) early.

■ Interview tip: the \$A\$1 vs A1 distinction is one of the most commonly tested Excel concepts in interviews — be ready to explain it out loud, and to explain the 'mixed reference' variants (\$A1 locks the column only, A\$1 locks the row only).

04 Formatting Cells

MEDIUM

Formatting doesn't change your data, but it changes how readable and trustworthy your report looks to a stakeholder — a polished-looking analysis gets taken more seriously.

CONCEPT / FUNCTION	WHAT IT DOES
Formatting	Changing appearance without changing the underlying value: currency, dates, alignment, bold, etc.
Format Painter	Copy formatting (not data) from one cell/range and apply it elsewhere instantly
Format Colors	Fill color and font color for emphasis or brand-consistent reports
Format Fonts	Font family, size, bold/italic/underline
Format Borders	Add lines around cells or ranges to visually separate sections
Format Numbers	Currency, percentage, decimal places, date formats — the display format, not the stored value
Format Grids / Settings	Gridlines visibility, freeze panes, and other view-level settings

Practical Example

A cell might STORE the value 0.156
but DISPLAY as 15.6% (Number Format: Percentage)

A cell might STORE 45832
but DISPLAY as \$45,832.00 (Number Format: Currency)
or as a date like 07/22/2026 (Number Format: Date)

The underlying number never changes - only how it's shown.

■ Data Analyst angle: consistent number formatting (especially currency, %, and dates) is often the difference between a report that looks 'client-ready' and one that looks like a rough draft — small effort, disproportionate impact on how your work is perceived.

■ Interview tip: be ready to explain that formatting is cosmetic — a cell formatted as currency still holds a plain number underneath, and formulas referencing it use the real number, not the displayed text.

05 Sorting & Filtering Data

HIGH

The Excel equivalent of SQL's *ORDER BY* and *WHERE*. This is often the very first thing you do with a raw dataset before any deeper analysis.

CONCEPT / FUNCTION	WHAT IT DOES
Sort	Reorder rows by one or more columns, ascending or descending — like SQL's <i>ORDER BY</i>
Custom Sort	Sort by multiple columns in a chosen priority order, or by a custom list (e.g. weekday names)
Filter	Show only rows matching a condition, hiding the rest without deleting data — like SQL's <i>WHERE</i>
AutoFilter Dropdowns	The small arrow icons in header cells that let you tick/untick values or set number/text conditions

Practical Example

Sort:

Select your data range -> Data tab -> Sort

Sort by: "Revenue", Order: Largest to Smallest

Filter:

Select your data range -> Data tab -> Filter

Click the dropdown arrow on "Region" -> untick all except "West", "East"

-> only West and East rows are now visible (others are hidden, not deleted)

■ Data Analyst angle: Sort + Filter is 80% of quick, everyday data exploration — before you ever build a PivotTable or write a formula, this is how you get a first feel for a new dataset.

■ Interview tip: point out that Filter hides rows, it does not delete or reorder the underlying data — a common interview question is 'what happens to hidden rows if I copy the visible range?' (answer: only visible rows are copied, if done correctly with Alt+; to select visible cells only).

06 Excel Tables

HIGH

Turning a plain range into a proper Excel Table unlocks structured references, automatic formula fill-down, and easy duplicate removal — a genuine step up from a 'range that looks like a table.'

CONCEPT / FUNCTION	WHAT IT DOES
Tables	Ctrl+T converts a range into a structured Table with named columns and auto-expanding rows
Table Design	Built-in styling, header row toggle, total row toggle, banded rows for readability
Table Resizing	Add new rows/columns and the Table automatically extends its range and formulas
Remove Duplicates	Built-in tool to instantly delete duplicate rows based on chosen columns — like SQL's DISTINCT, but destructive
Convert Table to Range	Turn a Table back into a plain range if you no longer need Table features
Table Style	Visual theme applied consistently across the whole Table

Practical Example

Convert a range to a Table:

Select data -> Ctrl+T -> confirm "My table has headers"

Now formulas can use structured references instead of cell addresses:

=SUM(Table1[Revenue])

instead of

=SUM(B2:B500)

Remove duplicates:

Table tab -> Remove Duplicates -> choose which columns define a "duplicate"

■ Data Analyst angle: Tables are what make PivotTables and charts auto-update when new rows are added — if you're rebuilding a monthly report from scratch every month, converting your source data to a Table is usually the fix.

■ Interview tip: mention structured references (Table1[ColumnName]) as a more readable, less error-prone alternative to raw cell ranges — this is a strong 'I know Excel beyond the basics' signal in interviews.

07 Conditional Formatting

MEDIUM

Turning numbers into visual signals — highlighting outliers, top performers, or values above/below a threshold at a glance, without a single formula being visibly written.

CONCEPT / FUNCTION	WHAT IT DOES
Conditional Formatting	Automatically apply formatting (color, icon, bar) to cells that meet a rule
Highlight Cell Rules	Greater Than, Less Than, Between, Equal To, Text That Contains, Date Occurring, Duplicate/Unique Values
Top/Bottom Rules	Highlight the top N or bottom N% of values in a range
Above/Below Average	Automatically highlight values relative to the range's own average
Data Bars	In-cell bar charts showing relative magnitude directly behind the number
Color Scales	Gradient coloring (e.g. red-to-green) to show a range of performance at a glance
Icon Sets	Small icons (arrows, traffic lights) added based on value thresholds
Manage Rules	View, edit, reorder, or delete all conditional formatting rules applied to a sheet

Practical Example

Highlight underperforming regions:
 Select the Revenue column -> Conditional Formatting -> Highlight Cell Rules
 -> Less Than -> type 45000 -> pick a red fill

Result: any cell below 45000 turns red automatically,
 and updates live if the underlying number changes.

■ Data Analyst angle: this is often the fastest way to make a raw table 'read itself' — a manager scanning a report with color-coded thresholds gets the story in 5 seconds instead of reading every number.

■ Interview tip: conditional formatting is **dynamic** — it recalculates automatically as data changes, unlike manually coloring cells by hand. Interviewers sometimes ask you to explain why that matters for a live dashboard.

08 Charts & Visualization

HIGH

Turning rows of numbers into a picture a non-technical stakeholder can understand in seconds — arguably the most visible, most judged skill in an analyst's toolkit.

CONCEPT / FUNCTION	WHAT IT DOES
Charts (Overview)	Insert tab -> Charts — Excel suggests chart types based on your selected data
Bar / Column Charts	Compare values across categories; bar = horizontal, column = vertical
Stacked Bar / Column	Show parts-of-a-whole across categories, stacked on top of each other
Pie Charts	Show proportion of a whole — best for a small number of categories only

CONCEPT / FUNCTION	WHAT IT DOES
Line Charts	Show trends over time — the standard choice for time-series data
Stacked Line / 100% Stacked	Show how multiple trends combine, or their relative share, over time
Radar Charts	Compare multiple variables for one or more items on a circular axis — more niche, used for multi-dimensional comparisons
Chart Customization	Titles, axis labels, legends, colors — turning a default chart into a presentation-ready one

Practical Example

Choosing the right chart:

- Comparing 5 regions' Q3 revenue -> Column Chart
- Showing revenue trend over 12 months -> Line Chart
- Showing market share of 4 competitors -> Pie Chart
- Showing revenue AND cost trend together -> Line Chart (two series)
- Showing budget category breakdown by department over time -> Stacked Column Chart

■ Data Analyst angle: this is often literally the deliverable a stakeholder sees — the chart, not the raw table. A correct analysis with a poorly chosen or messy chart still reads as a weak analysis to non-technical readers.

■ Interview tip: be ready to justify **why** you'd pick one chart type over another for a given dataset — 'because it looks nice' is a weak answer; 'because we're comparing parts of a whole across time' is a strong one.

09 PivotTables

HIGH

The single most powerful, most-interview-tested feature in Excel for analysts. A PivotTable summarizes thousands of rows into a clean cross-tab in seconds — no formulas required.

CONCEPT / FUNCTION	WHAT IT DOES
PivotTable	A drag-and-drop tool to summarize, group, and cross-tabulate large datasets instantly
Rows / Columns	Fields dragged here become the groupings of your summary (e.g. Region as rows, Month as columns)
Values	The field being aggregated — Sum, Average, Count, Max, Min, etc.
Filters	Restrict the whole PivotTable to a subset (e.g. only 'Active' customers)
Refresh	PivotTables don't auto-update when source data changes — you must manually refresh (or use a Table as the source)

Practical Example

Raw data: 10,000 rows of individual sales transactions
(Date, Region, Product, Salesperson, Revenue)

PivotTable setup:

Rows: Region

Columns: Quarter

Values: Sum of Revenue

Filter: Product = "Electronics"

Result: a clean 4x4 grid showing total Electronics revenue
per Region per Quarter - built in under a minute, no formulas.

■ Data Analyst angle: this is the Excel equivalent of SQL's GROUP BY + aggregate functions, and of pandas' groupby(). If you're comfortable with GROUP BY from the SQL notes, a PivotTable will feel immediately familiar — same idea, drag-and-drop interface.

■ Interview tip: PivotTables come up in almost every Excel-focused analyst interview — practice building one from a messy dataset live, and be ready to explain that they must be manually refreshed (Data -> Refresh All) after the source data changes.

10 Logical & Lookup Functions

HIGH

The functions that make Excel 'think' — applying business rules, flagging conditions, and pulling matching data from another table. VLOOKUP alone is one of the most interview-tested Excel skills that exists.

CONCEPT / FUNCTION	WHAT IT DOES
IF	=IF(condition, value_if_true, value_if_false) — the single most-used logical function in Excel
IFS	Test multiple conditions in one formula without nesting several IFs inside each other
AND	=AND(cond1, cond2, ...) — TRUE only if every condition is true
OR	=OR(cond1, cond2, ...) — TRUE if any condition is true
XOR	TRUE if an odd number of the given conditions are true — rarely used day-to-day, good to recognize
VLOOKUP	=VLOOKUP(value, table_range, column_index, exact_match) — look up a value in the first column of a range and return a value from another column in the same row

Practical Example

Business rule with IF:

=IF(B2>=50000, "Target Met", "Below Target")

Multiple conditions with AND:

=IF(AND(B2>=50000, C2="Active"), "Qualified", "Not Qualified")

VLOOKUP - pull a customer's country from a separate lookup table:
 =VLOOKUP(A2, CustomerTable!A:C, 3, FALSE)
 A2 -> the value to search for (customer ID)
 CustomerTable!A:C -> the range to search in
 3 -> return the value from the 3rd column of that range
 FALSE -> require an EXACT match (almost always what you want)

■ Data Analyst angle: VLOOKUP is the Excel equivalent of a SQL JOIN on a single column — if you're comfortable with LEFT JOIN from the SQL notes, VLOOKUP is that same 'match customer_id and pull their data' logic, just row-by-row in a spreadsheet.

■ Interview tip: always use FALSE (exact match) as VLOOKUP's last argument unless you specifically need approximate matching — using TRUE by accident is one of the most common real-world Excel bugs, and interviewers know to check for it.

11 Statistical & Math Functions

HIGH

The workhorse aggregate functions — the Excel equivalents of SQL's SUM/COUNT/AVG and the descriptive statistics you'd compute in pandas.

CONCEPT / FUNCTION	WHAT IT DOES
SUM	Adds all numbers in a range: =SUM(B2:B100)
SUMIF / SUMIFS	Sum only rows matching one condition (SUMIF) or several conditions (SUMIFS)
AVERAGE	Arithmetic mean of a range
AVERAGEIF / AVERAGEIFS	Average of only the rows matching one or more conditions
COUNT	Counts cells containing numbers only
COUNTA	Counts cells that are not empty (numbers, text, dates — anything)
COUNTBLANK	Counts empty cells in a range — useful for spotting missing data
COUNTIF / COUNTIFS	Count only rows matching one condition (COUNTIF) or several (COUNTIFS)
MAX / MIN	Largest / smallest value in a range
MEDIAN / MODE	The middle value / the most frequently occurring value in a range
STDEV.P / STDEV.S	Standard deviation for a full population (.P) vs. a sample (.S) — a common exact distinction interviewers test
NPV	Net Present Value — discounts future cash flows to today's value, common in finance-flavored analyst roles

CONCEPT / FUNCTION	WHAT IT DOES
RAND	Generates a random decimal between 0 and 1 — used for sampling or simulations

Practical Example

Basic aggregates:

`=SUM(C2:C500)`

`=AVERAGE(C2:C500)`

`=MAX(C2:C500)`

Conditional aggregates - the SQL "GROUP BY + WHERE" equivalent:

`=SUMIFS(C2:C500, B2:B500, "West", D2:D500, ">=45000")`

-> sum of Revenue (C) where Region (B) = "West" AND Revenue >= 45000

Data-quality check:

`=COUNTBLANK(B2:B500)` -> how many rows are missing a value in column B

■ Data Analyst angle: SUMIFS/COUNTIFS/AVERAGEIFS are your bread and butter for any KPI report — they are functionally identical to writing a SQL query with WHERE and an aggregate function, just inside a spreadsheet cell.

■ Interview tip: know the difference between STDEV.P (you have the entire population's data) and STDEV.S (you have a sample and are estimating the population) — this is a genuine statistics concept interviewers use to separate candidates who memorized functions from those who understand them.

12 Text Functions

MEDIUM

Excel's text-cleaning toolkit — smaller in scope than Python's string methods, but you'll reach for these constantly when raw data arrives messy inside a spreadsheet.

CONCEPT / FUNCTION	WHAT IT DOES
CONCAT	Joins multiple text values into one: <code>=CONCAT(A2, " ", B2)</code> combines first and last name
LEFT / RIGHT	Extract a set number of characters from the start (LEFT) or end (RIGHT) of a text string
LOWER	Converts text to lowercase — useful for standardizing inconsistent entries before comparison
TRIM	Removes extra spaces from text — the single most useful cleaning function for messy imported data

Practical Example

Combine first and last name:

`=CONCAT(A2, " ", B2)` -> "Amit Shah"

Extract a product code prefix:

`=LEFT(A2, 3)` -> first 3 characters, e.g. "ELE" from "ELE-4021"

Clean up inconsistent spacing from a data export:

`=TRIM(A2)` -> " John Smith " becomes "John Smith"

Standardize casing before comparing two columns:

`=LOWER(A2)=LOWER(B2)` -> TRUE/FALSE, ignoring case differences

■ Data Analyst angle: this is the direct Excel equivalent of the Python string-cleaning category in these notes — TRIM ≈ `.strip()`, LOWER ≈ `.lower()`, CONCAT ≈ f-strings/concatenation. Same job, different tool.

■ Interview tip: TRIM only removes extra *spaces*, not all whitespace characters (like line breaks) — for messier imports you may also need CLEAN(), which removes non-printable characters; mentioning this shows deeper hands-on experience.

Next Steps in Your Learning Journey

- **Practice on a real, messy dataset** — download any public CSV (sales, sports stats, survey data) and rebuild every category above on it: sort it, filter it, VLOOKUP against a second table, pivot it, chart it.
- **Connect the three tools you've now studied:** a PivotTable's Rows/Values = SQL's GROUP BY + aggregate function = pandas' groupby(). VLOOKUP = a SQL LEFT JOIN = pandas' merge(). Seeing the same idea three ways is what makes it stick for an interview.
- **Go beyond this list next:** INDEX/MATCH (a more flexible alternative to VLOOKUP), XLOOKUP if your Excel version has it, Power Query for repeatable data cleaning, and array formulas — these are common 'next level' topics in mid-level analyst interviews.
- **Speed matters:** practice keyboard shortcuts (Ctrl+T for Tables, F4 to toggle references, Alt+; to select visible cells only) — live Excel tests often reward speed as much as correctness.
- **Document the journey on LinkedIn:** a short post per category (e.g. 'Day 9: Finally understand why VLOOKUP needs FALSE as the last argument') builds a visible, credible trail for recruiters.

SQL, Python, and Excel are three different tools that all answer the same underlying questions: filter, group, join, summarize. Once you notice that pattern repeating across all three, you stop learning three separate skills and start recognizing one skill wearing three outfits — that's when interview prep gets fast.